



Continuous Integration with CodeBuild



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```
! buildspec.yml
1  version: 0.2
2
3  phases:
4    install:
5      runtime-versions:
6        java: corretto8
7    pre_build:
8      commands:
9        - echo Initializing environment
10       - export CODEARTIFACT_AUTH_TOKEN=`aws codeartifact get-authorization-token --domain nextwork --domain-rol
11
12    build:
13      commands:
14        - echo Build started on `date`
15        - mvn -s settings.xml compile
16    post_build:
17      commands:
18        - echo Build completed on `date`
19        - mvn -s settings.xml package
20  artifacts:
21    files:
22      - target/nextwork-web-project.war
23    discard-paths: no
24
```



Introducing Today's Project!

'In this project, I will demonstrate how to use AWS codebuild to automate the build process in the CI/CD pipeline. Building means compressing our webapp files into a single compressed file that can be deployed later.

Key tools and concepts

key Services I used were Codebuild, Code connections, Code artifact, Github. Key concepts I learnt include the build process, resolving git issues in the terminal and using code connections to connect github and Aws

Project reflection

This project took me approximately 3 hours to complete. The most challenging part was resolving the git permission issues. It was most rewarding to see the build success at the end.

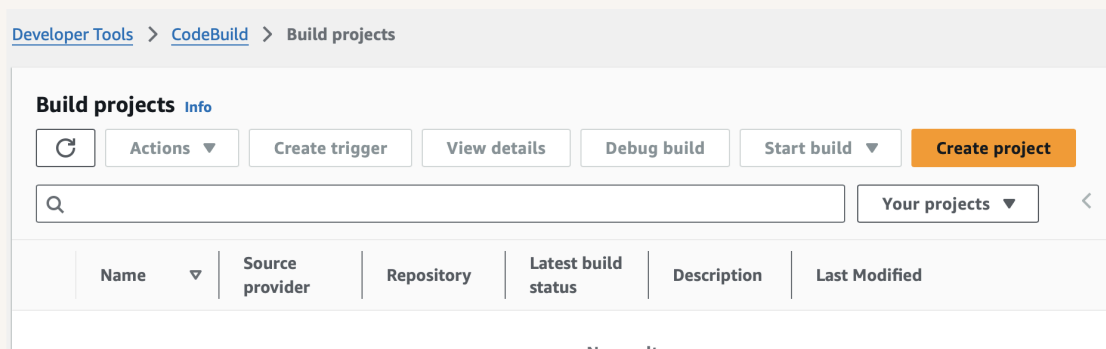
This project is part four of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow



Setting up a CodeBuild Project

CodeBuild is a continuous integration service , which means it runs tests, compiles code, and makes sure new changes play nicely with the existing codebase.

My CodeBuild project's source means the location of the code that CodeBuild will fetch, compile, and package into a file you can deploy, and I selected Github





Connecting CodeBuild with GitHub

There are multiple credential types for GitHub, like personal access tokens and OAuth app. I used GitHub App because AWS manages the application and connection, reducing the need for me to handle tokens or keys directly.

The service that helped connect us is AWS CodeConnections it's like a secure bridge between AWS and your external code repositories.

Source 1 - Primary

Source provider

GitHub ▼

Credential

✔ Your account is successfully connected by using an AWS managed GitHub App.
[Manage account credentials.](#)

☐ Use override credentials for this project only

Repository

☒ Repository in my GitHub account

☐ Public repository



CodeBuild Configurations

Environment

My CodeBuild project's Environment configuration means the setup for the ec2 instance that will be compiling and compressing the files when we run our code build project. it includes settings like provisioning model, compute, operating system etc.

Artifacts

Build artifacts are files that are created as a part of the code build process . My build process will create a compressed file representing the webapp to be deployed later on. I created an S3 bucket to store them

Packaging

When setting up CodeBuild, I also chose to package artifacts into a zip file because it helps with package management as files are all organized neatly in a single executable file.



Monitoring

For monitoring, I enabled CloudWatch Logs, which collects and tracks logs from AWS services. In this project, CloudWatch will record everything that happens during the build process, including the commands that are run and any errors that occur.



buildspec.yml

My first build failed because my buildspec.yml file was absent in my github repo. A buildspec.yml file is needed because CodeBuild reads the buildspec.yml file like a step-by-step instruction manual. It goes through each phase in order to run commands

The first two phases in my buildspec.yml file are tasks to do before the main building starts. The third phase is where the actual building happens. The fourth phase are the finishing touches after the main build is done.

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21    files:
22      - target/nextwork-web-project.war
23    discard-paths: no
24
```



Success!

My second build also failed, but with a different error that said `COMMAND_EXECUTION_ERROR: Error while executing command: mvn -s settings.xml compile`. To fix this we need to grant CodeBuild's IAM role the permission to access CodeArtifact.

To resolve the second error, I added the code artifact permissions to my code build service role. When I built my project again, I got the build successful prompt.

To verify the build, I checked my S3 bucket. Seeing the `nextwork-web-project.war` file inside the downloaded zip archive, confirmed the build process produced the expected output.

[Developer Tools](#) > [CodeBuild](#) > [Build projects](#) > [nextwork-devops-cicd](#) > [nextwork-devops-cicd:eb81f90a-a84a-496d-b...](#)

nextwork-devops-cicd:eb81f90a-a84a-496d-ba8d-b10e42c000ca

Stop build

Debug build

Retry build

Build status

Status	Initiator	Build ARN
✔ Succeeded	Tunechie_projects	arn:aws:codebuild:us-east-1:891377395566:build/nextwork-devops-cicd:eb81f90a-a84a-496d-ba8d-b10e42c000ca
Resolved source version	Start time	End time
4f13863705335364aac49c0a75b3be5c0a3030c8	Apr 18, 2025 3:59 PM (UTC-4:00)	Apr 18, 2025 4:00 PM (UTC-4:00)



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